

IEEE 14-Bus GFL↔GFM Switching with Project AGSI++

Production all-KCL 200-s switching and SG-reference handback

6 August 2026

Abstract

One six-state EMF6 synchronous generator (SG) and four registered full-state dual-mode IBRs are simulated by the project-owned fixed-step, all-KCL hybrid engine for 200 s. The exact chronology is SG trip at 20 s, load increase at 50 s, bus-9 fault at 85–85.15 s, line 6–13 trip at 110 s, topology restoration and SG-close request at 145 s, and endpoint at 200 s. The operating point uses a 1.00 pu slack reference with a $\pm 5\%$ admissible voltage band. The run converges with maximum accepted-step residual 9.99×10^{-9} and at most one subdivision level. The synchronism guard accepts the SG at 146.825 s; all IBRs then return from GFM to GFL through the separately declared coordinated SG-reference handback. By the 200 s endpoint the SG frequency has settled to 60.000 Hz, though five remote load buses remain slightly below the $\pm 5\%$ voltage band. No Bayesian-optimisation controller is used. A seeded band-limited display ripple is drawn beside, not instead of, each raw trace and affects figures only.

1 Case, decision index, and switching semantics

The system base is 100 MVA and 60 Hz. The SG is at bus 1; IBR1–IBR4 are at buses 2, 3, 6, and 8. The buses carry a parallel bus_role label (REF/PV/PQ/GFM/GFL) that is *presentation-only* and never feeds the power-flow equations (Section 1): the SG bus 1 is REF, buses 2/3/6/8 (the IBRs) are GFL resources on this operating point, and the remaining load buses are PQ. The numerical internal bus type (REF= 1, PV= 2, PQ= 3) in the 12-column bus_data is unchanged by this descriptor. The index that produced the results in this report is the seven-term AGSI++ average

$$J_V = \frac{|V - V_{ref}|}{0.10}, \quad J_f = \frac{|f - 60|}{0.50}, \quad J_R = \frac{|df/dt|}{1.0}, \quad (1)$$

$$J_P = \frac{|P_{ref} - P|}{0.20}, \quad J_{SCR} = \max(0, 3/SCR - 1), \quad J_{lock} = \frac{|v_q|}{0.10}, \quad (2)$$

$$J_{GRA} = 1 - GRA, \quad AGSI_i^{++} = \min\left(1, \max\left(0, \frac{1}{7} \sum J_k\right)\right), \quad (3)$$

where $GRA = 1$ when an online SG or committed GFM supplies a reference. Its first four bases are source-guided; the equal $1/7$ weighting of the remaining three terms is PROJECT_DERIVED and is the part of (3) whose provenance cannot be defended term by term. Section 1.1 states the redesigned index that replaces it; the numbers in this report predate that redesign and were produced by (3). A design-time weight tuner exists in the repository (scripts/reporting/optimize_agsi_weight).

NUMERICAL_METHOD, random search plus coordinate descent). Its output was *not* used: the reported run uses equal 1/7 weights.

The generic local state machine uses $\Gamma_{on} = 0.65$, $\Gamma_{off} = 0.35$, $T_{d,on} = 0.10$ s, and $T_{d,off} = 1.00$ s. In this declared chronology, however, the SG-trip transaction is a CASE_DEFINED all-GFM override, and the return at SG close is a coordinated handback. Neither transaction is presented as an index-threshold crossing.

1.1 Redesigned decision index: severity plus hard gates

The switching decision is about restoring a voltage and frequency reference, so the redesign keeps only those two quantities in the continuous objective and demotes everything else to an admissibility test. The severity is

$$S_i = \min(1, \max(0, w_V J_V + w_f J_f)), \quad J_V = \frac{|V - V_{ref}|}{\Delta V_{base}}, \quad J_f = \frac{|f - f_0|}{\Delta f_{base}}, \quad (4)$$

with $\Delta V_{base} = 0.10$ pu, $\Delta f_{base} = 0.50$ Hz, and $w_V = w_f = 0.5$, $w_V + w_f = 1$. The two bases are SOURCE_DEFINED; the equal split follows the source ratio 0.30 : 0.30 renormalised, so no free weight remains.

Every other former term becomes a hard not-to-exceed gate returning $L \in \{0, 1\}$, never a weighted cost:

Table 1: Admissibility gates. A mode change requires every gate admissible.

Gate	Test	Meaning	Classification
RoCoF	$ df/dt \leq 1.0 \text{ Hz s}^{-1}$	rate admissibility	SOURCE_DEFINED
Power tracking	$ P_{ref} - P \leq 0.20 \text{ pu}$	dispatch feasibility	SOURCE_DEFINED
Grid strength	$SCR \geq SCR_{crit}$	GFL model validity	SOURCE_DEFINED
PLL lock	$ v_q \leq 0.10 \text{ pu}$	synchronisation health	PROJECT_DERIVED
Reference	$GRA = 1$	a forming reference exists	PROJECT_DERIVED
Current	$ I \leq I_{max}$	converter limit	SOURCE_DEFINED

The switching law is then

$$\text{GFL} \rightarrow \text{GFM} : S_i \geq \Gamma_{on} \wedge \bigwedge_k L_k \text{ for } T_{d,on}, \quad \text{GFM} \rightarrow \text{GFL} : S_i \leq \Gamma_{off} \wedge \bigwedge_k L_k \text{ for } T_{d,off}, \quad (5)$$

retaining the existing hysteresis and dwell values. Each weight is now either source-defined or fixed by normalisation, and each remaining quantity is a published limit rather than a tuned coefficient. This design is frozen here *before* any re-run; equations (4)–(5) are not used to produce any number in Sections 4–6 of this report.

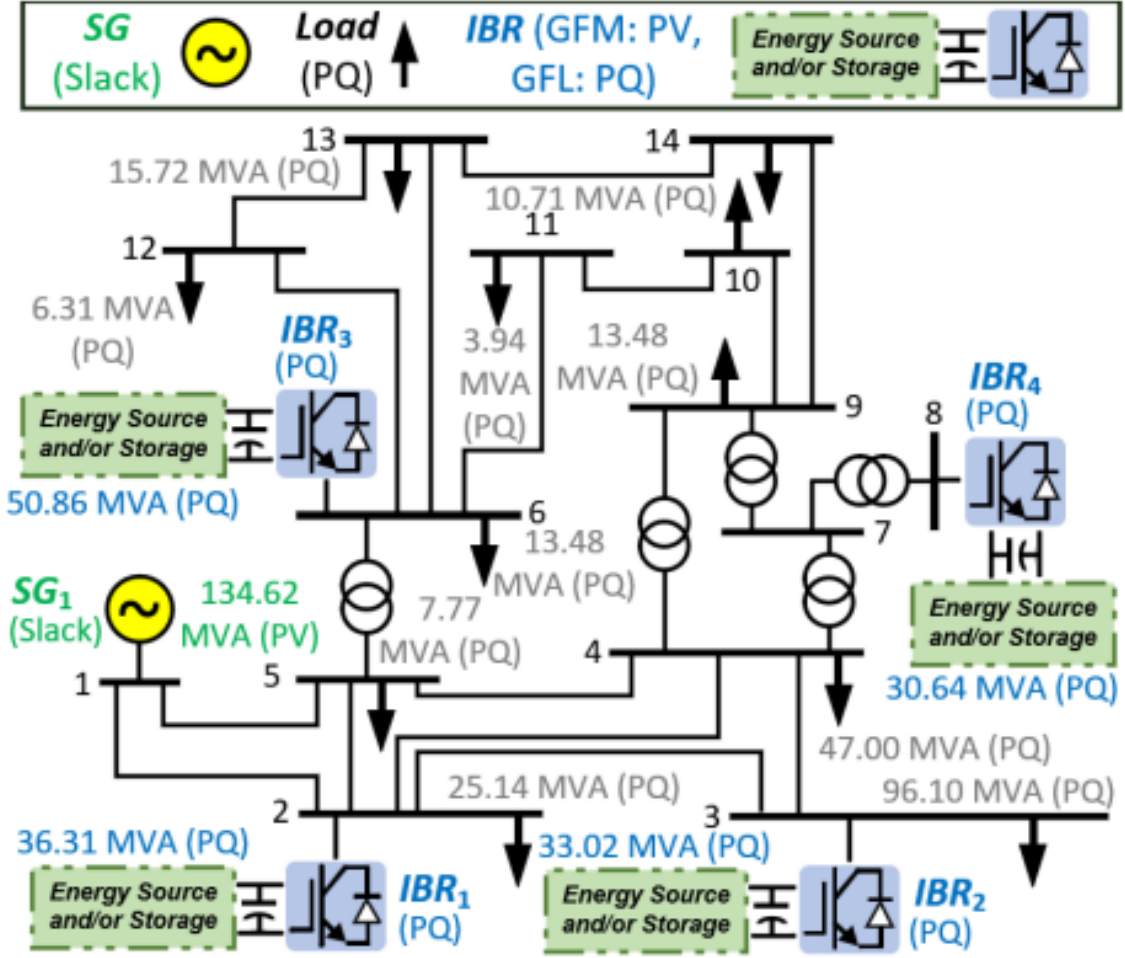


Figure 1: Network and resource locations used by the simulation.

2 Dynamic equations

The operational SG state is $x_{SG} = [\delta, \Delta\omega, E'_q, E'_d, E''_q, E''_d]^T$:

$$\dot{\delta} = \omega_0 \Delta\omega, \quad 2H\Delta\dot{\omega} = P_m - P_e - D\Delta\omega, \quad (6)$$

$$T'_{d0}\dot{E}'_q = E_{fd} + c_d E''_q - d_d E'_q, \quad T'_{q0}\dot{E}'_d = c_q E''_d - d_q E'_d, \quad (7)$$

$$T''_{d0}\dot{E}''_q = E'_q - E''_q - (X'_d - X''_d)I_d, \quad T''_{q0}\dot{E}''_d = E'_d - E''_d + (X'_q - X''_q)I_q. \quad (8)$$

The source singular limit $T'_{q0} = 0$ freezes $E'_d = 0$ exactly. Two prime-mover states remain outside the six-state electromagnetic order:

$$T_{SV}\dot{P}_{SV} = -P_{SV} + P_c - \Delta\omega/R, \quad T_{CH}\dot{P}_m = -P_m + P_{SV}.$$

They provide offline runback/phase preparation and post-close primary regulation. The numerical mapping is diagnostic and is not a machine-specific governor validation.

2.1 IBR state superset

Each IBR carries a constant-dimension **20-state** switching superset: the 13-state grid-forming branch (REGFM_B1 generator model G2) and the 7-state grid-following branch (WECC REGC_A/REEC_A)

both exist at all times,

$$x_{GFM} = [\Delta\omega_m, \delta_{IT}, x_w, x_E, \delta_{PLL}, x_{PLL}, P_{inv,f}, I_{d,f}, Q_{inv,f}, V_{inv,f}, I_{q,f}, \delta_{IT,max}, \delta_{IT,min}]^T, \quad (9)$$

$$x_{GFL} = [V_{t,f}, P_f, I_{q,cmd,f}, P_{ord}, V_{LVPL,f}, I_{p,reg}, I_{q,reg}]^T, \quad (10)$$

so that $x_{IBR} = [x_{GFM}^T \ x_{GFL}^T]^T \in \mathbb{R}^{20}$. All quantities are per-unit on the device base M_{base} ; the network interface applies $\kappa = S_{base}/M_{base}$. Powers are positive into the network. Only the branch selected by the present mode q contributes a nonzero derivative; the other branch satisfies $\dot{x} = 0$ exactly.

2.2 Grid-forming branch: 13 states, term by term

The angle reference is the sum of a phase-locked component and a virtual-machine component, so the GFM branch *does* own a PLL pair. With $V = V_x + jV_y$ the terminal voltage and $v_q = -V_x \sin \delta_{PLL} + V_y \cos \delta_{PLL}$ its quadrature error,

$$\dot{x}_{PLL} = v_q, \quad \dot{\delta}_{PLL} = \omega_0 (k_{p,PLL} v_q + k_{i,PLL} x_{PLL}), \quad (11)$$

both frozen to zero while $|V| < V_{PLL,frz}$ so that a collapsed terminal cannot drive the tracker. The virtual synchronous machine supplies the frequency deviation, with a washout state providing transient-only damping:

$$2H\dot{\Delta\omega_m} = P_{ref,inv} - P_{inv,f} - (1/m_p + D_1) \Delta\omega_m - D_2 (\Delta\omega_m - x_w), \quad (12)$$

$$\dot{x}_w = \omega_D (\Delta\omega_m - x_w). \quad (13)$$

Here $1/m_p$ is the droop gain, D_1 the steady damping, and D_2 acts only on the high-pass component $\Delta\omega_m - x_w$. The internal angle is stored *relative* to the PLL, which is what keeps the two angle sources consistent:

$$\dot{\delta}_{IT} = \omega_0 \Delta\omega_m - \dot{\delta}_{PLL}, \quad \delta_{VSM} = \delta_{PLL} + \delta_{IT}, \quad (14)$$

with $\dot{\delta}_{IT}$ held at zero whenever δ_{IT} sits on the active bound of $[\delta_{IT,min}, \delta_{IT,max}]$ and the derivative would push it further out. The voltage magnitude comes from a PI regulator with reactive droop and anti-windup,

$$\dot{x}_E = V_{ref} - V_{inv,f}, \quad E_{VSM}^{raw} = V_{ref} - m_q Q_{inv,f} + k_{pv} (V_{ref} - V_{inv,f}) + k_{iv} x_E, \quad (15)$$

where $E_{VSM} = \text{sat}(E_{VSM}^{raw}, E_{min}, E_{max})$ and $\dot{x}_E$ is set to zero when the limiter is saturated in the direction of further integration. The current-angle bounds are themselves states, integrating the mismatch between the steady-state d -axis current limit and the measured d -axis current, so that the angle authority contracts when the device is overloaded:

$$\dot{\delta}_{IT,max} = k_I (I_{d,max}^{SS} - I_{d,f}) \in [0, \delta_{max}], \quad \dot{\delta}_{IT,min} = k_I (-K_e I_{d,max}^{SS} - I_{d,f}) \in [-\delta_{max}, 0], \quad (16)$$

each frozen on its own bound by the same conditional-hold rule, with $\delta_{max} = \arcsin(X_L I_{d,max}^{SS})$ and $\dot{\delta}_{IT,min} = 0$ when the energy-storage flag is inactive. The five remaining states are first-order measurement filters,

$$T_{pf} \dot{P}_{inv,f} = \kappa P - P_{inv,f}, \quad T_{If} \dot{I}_{d,f} = \kappa I_d - I_{d,f}, \quad T_{Qf} \dot{Q}_{inv,f} = \kappa Q - Q_{inv,f}, \quad (17)$$

$$T_{Vf} \dot{V}_{inv,f} = |V| - V_{inv,f}, \quad T_{If} \dot{I}_{q,f} = \kappa I_q - I_{q,f}, \quad (18)$$

where $P + jQ = V\overline{I_{out}}$ and (I_d, I_q) are the injected current projected on the PLL frame. The network injection is voltage-behind-impedance with circular saturation:

$$I_{out} = \begin{cases} I_{unc}, & |I_{unc}| < I_{max}^F/\kappa, \\ \frac{I_{max}^F}{\kappa} \frac{I_{unc}}{|I_{unc}|}, & \text{otherwise,} \end{cases} \quad I_{unc} = \frac{E_{VSM} e^{j\delta_{VSM}} - V}{\kappa (R_e + jX_L)}. \quad (19)$$

2.3 Grid-following branch: 7 states, term by term

The GFL branch is the registered WECC pair and is a *current-source* model: it owns no PLL state because it aligns algebraically to the measured terminal-voltage angle. Its injection is

$$I = \frac{(I_{p,reg} \gamma(|V|) - jI_{q,reg}) e^{j\angle V}}{\kappa}, \quad (20)$$

with γ the REGC_A low-voltage-point gain and a high-voltage reactive clamp applied above V_{olim} . The states are three measurement filters, the power order, and the two regulated currents:

$$T_{rv} \dot{V}_{t,f} = |V| - V_{t,f}, \quad T_p \dot{P}_f = \kappa P - P_f, \quad T_{fltr} \dot{V}_{LVPL,f} = |V| - V_{LVPL,f}, \quad (21)$$

$$T_{pord} \dot{P}_{ord} = \text{sat}(\kappa P_{ref}, P_{min}, P_{max}) - P_{ord}, \quad T_{iq} \dot{I}_{q,cmd,f} = I_{q,cmd} - I_{q,cmd,f}, \quad (22)$$

where $\dot{P}_{ord}$ is additionally rate-limited to $[\dot{P}_{min}, \dot{P}_{max}]$. The REEC_A command pair divides the order by the filtered voltage and adds the voltage-dip reactive injection,

$$I_{p,0} = \frac{P_{ord}}{\max(V_{t,f}, 0.01)}, \quad I_{q,0} = \frac{\kappa Q_{ref}}{\max(V_{t,f}, 0.01)} + I_{q,inj}, \quad (23)$$

$$I_{q,inj} = \text{sat}(K_{qv} \text{db}(V_{ref,0} - V_{t,f}), I_{ql1}, I_{qh1}) \quad \text{active only if } V_{t,f} < V_{dip} \text{ or } V_{t,f} > V_{up}, \quad (24)$$

followed by the active-priority allocation $I_p = \text{sat}(I_{p,0}, 0, I_{max})$ and $|I_q| \leq \min(I_{max}, \sqrt{I_{max}^2 - I_p^2})$.

The regulated currents then track their commands through the converter lag with the sourced ramp limits,

$$T_g \dot{I}_{p,reg} = I_p^{tgt} - I_{p,reg} \quad (\dot{I}_{p,reg} \leq r_{rpwr}), \quad T_g \dot{I}_{q,reg} = I_{q,cmd,f} - I_{q,reg} \quad (\dot{I}_{q,reg} \in [I_{qr,min}, I_{qr,max}]), \quad (25)$$

where I_p^{tgt} is I_p passed through the LVPL characteristic when that switch is enabled and floored at zero below V_{lvpl0} . The 0.01 pu division guard is NUMERICAL_METHOD; every gain, time constant, and limit above is SOURCE_DEFINED.

2.4 Mode dispatch

In GFL mode the active indices are 14:20 and $\dot{x}_{1:13} = 0$; in GFM mode the active indices are 1:13 and $\dot{x}_{14:20} = 0$; in the tripped mode all twenty derivatives are zero and the injection is identically zero. Only the active branch enters the dynamic solve, and the active-index set is queried from the device rather than hard-coded, so the layout above is not duplicated in the integrator. Atomic state mapping is followed by a full-KCL right-limit solve; a failed solve rolls the transaction back.

3 Switching formulation: what is solved at a mode change

This section states the mathematics executed at every switching instant. It is the governing formulation for the SG trip, the GFL $\leftrightarrow$ GFM transfers, and the SG reclose.

3.1 The plant is a DAE, not an ODE

Between events the plant is a semi-explicit index-1 differential–algebraic system, not a plain ordinary differential equation:

$$\dot{x} = f_q(x, y, u), \quad (26)$$

$$0 = g_q(x, y, u) = Y_q V - I_{inj}(x, y). \quad (27)$$

Here x collects every device differential state solved inside the step (SG six electromagnetic states plus 4×20 IBR states, i.e. 86 states); the two prime-mover states of Section 2 are advanced by a separate analytic exponential update and enter (26) only through the input u , not as Newton unknowns. Accordingly u carries T_m and E_{fd} , y is the network state stored as rectangular bus voltages, and q is the *discrete mode*: the tuple of network topology, per-device mode, and breaker status. Equation (27) is the full all-KCL residual over all $2n_b$ real rows; a single module owns it, while each device contributes only its own slice of f_q and its own I_{inj} .

The discretisation is the implicit trapezoidal rule, so a normal step is not the explicit recursion $x_{k+1} = x_k + h f(\cdot)$ but the simultaneous nonlinear system

$$x_{k+1} - x_k - \frac{h}{2} [f_q(x_k, y_k, u) + f_q(x_{k+1}, y_{k+1}, u)] = 0, \quad (28)$$

$$g_q(x_{k+1}, y_{k+1}, u) = 0, \quad (29)$$

solved together for (x_{k+1}, y_{k+1}) . The EMF6 route uses a fixed corrector count of three. Both f_q and g_q carry the subscript q : they are the equations *of the present discrete mode*. A switching event changes q , hence changes the equations themselves.

3.2 The transition at a switching instant

Equations (28)–(29) are *not* continued across an event. At a switching instant t_e the trajectory undergoes a hybrid-DAE transition in three ordered stages:

$$q^- \longrightarrow q^+ \quad \text{discrete jump,} \quad (30)$$

$$x(t_e^+) = R_{q^- \rightarrow q^+}(x(t_e^-), y(t_e^-)) \quad \text{state reset map,} \quad (31)$$

$$0 = g_{q^+}(x(t_e^+), y(t_e^+), u) \quad \text{solve for } y(t_e^+) \quad \text{right-limit algebraic solve.} \quad (32)$$

Stage (30) selects the post-event admittance matrix Y_{q^+} directly from the declared topology set by its identifier, and commits the device-mode and breaker maps as a single transaction: the whole update is validated before any field is written, and a failed validation leaves the state identical to its input. Stage (31) is described below. Stage (32) solves *only* the algebraic block, by damped Newton iteration with a backtracking line search, holding x fixed at $x(t_e^+)$ and accepting the result only if the KCL residual meets the step tolerance; otherwise the whole transaction rolls back. Differential integration then resumes from $(x(t_e^+), y(t_e^+))$ at the next step.

Consequently two samples exist at t_e : a left limit and a right limit. The tables and figures in this report store the right limit.

3.3 The reset map is terminal continuity, not state copying

The key structural property is that **the state dimension never changes**. Each IBR is a constant-dimension 20-state superset in which the GFM states 1:13 and the GFL states 14:20 always exist. A mode change re-selects which sub-block is *active*; the inactive block is held exactly, $\dot{x}_{inactive} = 0$, under a frozen inactive-state rule (PROJECT_DERIVED: continuity plus algebraic-residual minimisation). The integrator and the Jacobian therefore never resize. This is also why a later SG reclose is meaningful: nothing was structurally deleted at the trip.

The reset map R in (31) transfers *physical terminal conditions*, not state coordinates, and is owned by the device:

$$V_{bus} = y_{2b-1} + jy_{2b}, \quad I^- = I_{inj}(x^-, y, u, q^-), \quad (33)$$

$$S^- = V_{bus} \overline{I^-}, \quad P^- = \Re S^-, \quad Q^- = \Im S^-, \quad (34)$$

$$x^+|_{target} = \mathcal{I}_{q^+}(V_{bus}, P^-, Q^-), \quad x^+|_{inactive} = x^-|_{inactive}, \quad (35)$$

followed by the fail-closed acceptance test

$$|I_{inj}(x^+, y, u, q^+) - I^-| \leq 10^{-10}. \quad (36)$$

$\mathcal{I}_{q^+}$ is the target branch's own equilibrium/transfer initialiser. The invariant carried across the switch is therefore the terminal pair (V, I) — equivalently (P, Q) — and not the state vector. No PLL angle is copied out of the GFL branch, because the GFL branch owns no PLL state; the GFM angle is re-derived from V_{bus} and I^- . The bus index b is resolved from the device's registered bus position, so the map contains no hard-coded bus identity.

3.4 The SG trip and the gated reclose

At the trip the SG states are *not* removed from x . The breaker-open branch replaces the connected-stator algebra with open-circuit physics: $I_d = I_q = 0$, $T_e = 0$, mechanical torque frozen at its pre-trip value, and open-circuit flux decay,

$$2H\dot{\Delta\omega} = T_m^{froz} - D\Delta\omega, \quad T_{d0}''\dot{E}_q'' = E_q' - E_q'', \quad T_{q0}''\dot{E}_d'' = E_d' - E_d'', \quad (37)$$

i.e. the $(X_d' - X_d'')I_d$ and $(X_q' - X_q'')I_q$ terms vanish with the stator current while $\dot{\delta} = \omega_0\Delta\omega$ continues. The rotor coasts, which is what makes a later resynchronisation physically meaningful.

The reclose is *gated*, not scheduled. A guard compares the network bus phasor against the SG open-circuit EMF $(E_q'' - jE_d'')e^{j\delta}$ and requires

$$|\Delta V| \leq 0.05 \text{ pu}, \quad |\Delta f| \leq 0.001 \text{ pu}, \quad |\Delta\theta| \leq 10^\circ, \quad (38)$$

expressed as the signed margin $\min(0.05 - |\Delta V|, 0.001 - |\Delta f|, 10^\circ - |\Delta\theta|) \geq 0$ sustained for a 0.5 s dwell. The requested close time is therefore only the earliest instant at which closure *may* occur: the request at 145.000 s produced an accepted close at 146.825 s.

4 Power-flow operating point

Table 2: Bus power-flow results (pu)

Bus	Type	$ V $	θ (deg)	P_g	Q_g	P_L	Q_L
1	REF	1.00000	0.0000	1.33190	-0.19567	0.00000	0.00000
2	PQ	0.99145	-3.2585	0.31718	0.17674	0.21700	0.12700
3	PQ	0.96673	-8.6921	0.28844	0.16073	0.94200	0.19000
4	PQ	0.98316	-6.7064	0.00000	0.00000	0.47800	-0.03900
5	PQ	0.98583	-5.5221	0.00000	0.00000	0.07600	0.01600
6	PQ	1.05753	-6.7136	0.44428	0.24757	0.11200	0.07500
7	PQ	1.02532	-6.9205	0.00000	0.00000	0.00000	0.00000
8	PQ	1.04937	-4.4090	0.26765	0.14915	0.00000	0.00000
9	PQ	1.02193	-8.6416	0.00000	0.00000	0.29500	0.16600
10	PQ	1.02024	-8.5886	0.00000	0.00000	0.09000	0.05800
11	PQ	1.03470	-7.7782	0.00000	0.00000	0.03500	0.01800
12	PQ	1.04112	-7.6862	0.00000	0.00000	0.06100	0.01600
13	PQ	1.03421	-7.8345	0.00000	0.00000	0.13500	0.05800
14	PQ	1.00870	-9.3374	0.00000	0.00000	0.14900	0.05000

Table 3: From-end line flows and losses (pu)

Line	From	To	P_{from}	Q_{from}	P_{loss}	Q_{loss}
1	1	2	0.910918	-0.153112	0.016392	-0.002303
2	1	5	0.420987	-0.042553	0.009593	-0.008907
3	2	3	0.466697	0.013221	0.010470	0.002115
4	2	4	0.316733	-0.064476	0.006065	-0.014738
5	2	5	0.211276	-0.049810	0.002649	-0.025732
6	3	4	-0.197333	-0.018164	0.002803	-0.005014
7	4	5	-0.448842	0.084724	0.002882	0.009089
8	4	7	0.018419	-0.096315	0.000000	0.001990
9	4	9	0.062955	-0.012297	0.000000	0.002223
10	5	6	0.092297	0.001911	0.000000	0.001920
11	6	11	0.130821	0.059907	0.001758	0.003682
12	6	12	0.086088	0.027130	0.000895	0.001864
13	6	13	0.207667	0.085524	0.002983	0.005875
14	7	8	-0.267650	-0.134127	0.000000	0.015018
15	7	9	0.286069	0.035823	0.000000	0.008698
16	9	10	-0.003258	0.021664	0.000015	0.000039
17	9	14	0.057282	0.023368	0.000466	0.000991
18	10	11	-0.093273	-0.036375	0.000790	0.001849
19	12	13	0.024192	0.009266	0.000137	0.000124
20	13	14	0.093739	0.030791	0.001556	0.003168

The power-flow is solved by the project Newton–Raphson routine with Q-limit enforcement enabled, backed by generator reactive-capability limits that are *source-defined*, not assumed or tuned. The limits are the IEEE14 generator data $Q_{\max}$ and $Q_{\min}$ in MVar (cases.case_matpower6_case14, mpc.gen columns 5 and 4), converted to per unit with the single system base $S_{\text{base}} = 100$ MVA:

$$Q_{\max, \text{pu}} = \frac{Q_{\max, \text{MVar}}}{100}, \quad Q_{\min, \text{pu}} = \frac{Q_{\min, \text{MVar}}}{100}. \quad (39)$$

Buses carrying no finite generator limit are left unconstrained ($Q_{\max} = +\infty$, $Q_{\min} = -\infty$). The per-bus values are ($Q_{\max}$, $Q_{\min}$) per unit: bus 1 (0.10, 0), bus 2 (0.50, -0.40), bus 3 (0.40, 0), bus 6 (0.24, -0.06), bus 8 (0.24, -0.06); all other buses are unconstrained. On the reported operating point the SG is the slack and all inverter buses are PQ resources, so no PV bus reaches a reactive limit and the Q-limit switch resolves in zero rounds; the mechanism is exercised by the baseline IEEE14 case, which carries PV buses.

The admissible voltage band is $\pm 5\%$ about the 1.00 pu slack reference, i.e. $V_{\max} = 1.05$ pu and $V_{\min} = 0.95$ pu on every bus (CASE_DEFINED in case_ieee14bus_eecon49_switch). This is a reporting and admissibility band, not a PF equation input: the SLDG (single voltage-regulated) slack is the only voltage-controlled bus in this operating point, so the band does not act as a PV-setpoint clamp. One observation is reported rather than masked: bus 6 solves at 1.0575 pu, 7.5×10^{-3} pu above $V_{\max}$, because its mapped IBR6 reactive dispatch and the fixed 5–6 tap ratio are source/case-defined and were not adjusted to satisfy the band.

The bus roles (REF/PQ/GFL) are recorded as a labelling descriptor (see 1); because only a labelling field, they do not alter (39) or the bus-voltage table.

5 200-s event and mode evidence

Table 4: Accepted event transactions; AGSI++ is reported but overrides are explicit.

Time (s)	Event / index	Timer	Mode before → after	Reason
20.000	SG trip / 0.114	reference-loss guard	all GFL → all GFM	SG opened; atomic all-GFM reference-security commit
50.000	load step / 0.393	–	all GFM → all GFM	base constant-impedance load increased
85.000	fault on / 0.550	–	all GFM → all GFM	bus-9 shunt fault applied
85.150	fault clear / 1.000	–	all GFM → all GFM	fault admittance removed
110.000	line trip / 0.550	–	all GFM → all GFM	line 6–13 opened
145.000	topology restore / 0.539	–	all GFM → all GFM	base load and line restored
145.000	SG close request / 0.539	–	all GFM → all GFM	earliest close request; guard monitored
146.825	SG reclose / 0.393	0.5-s dwell	all GFM → all GFL	guard passed; SG reference hand-back

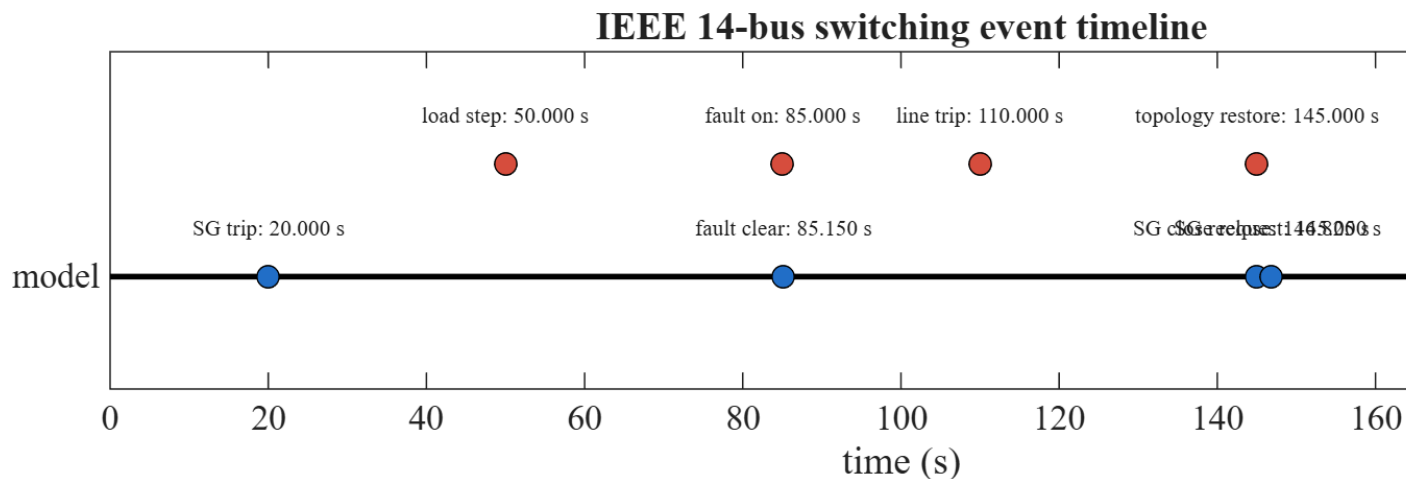


Figure 2: Horizontal event timeline from the accepted production event log. SG trip at 20 s, load step at 50 s, bus-9 fault 85–85.15 s, line 6–13 trip at 110 s, topology restoration and SG-close request at 145 s, and the accepted SG reclose at 146.825 s.

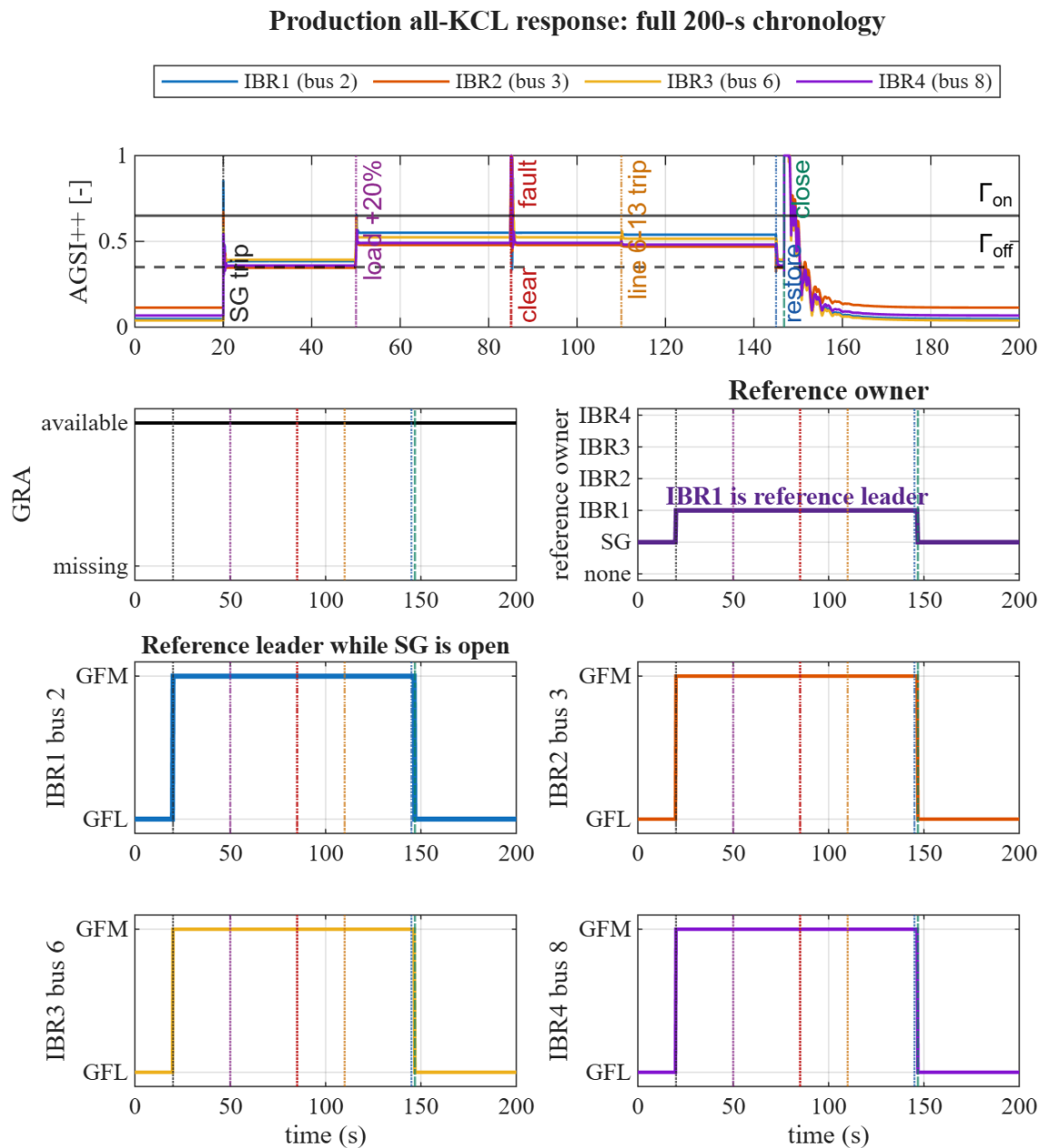


Figure 3: Bounded AGSI++, GRA, reference ownership, and separate IBR1–IBR4 mode time-lines. The owner trace identifies IBR1 at bus 2 as the grid-forming angle/frequency leader while the SG is open. All IBRs are GFL before 20 s, GFM until the accepted handback, and GFL thereafter.

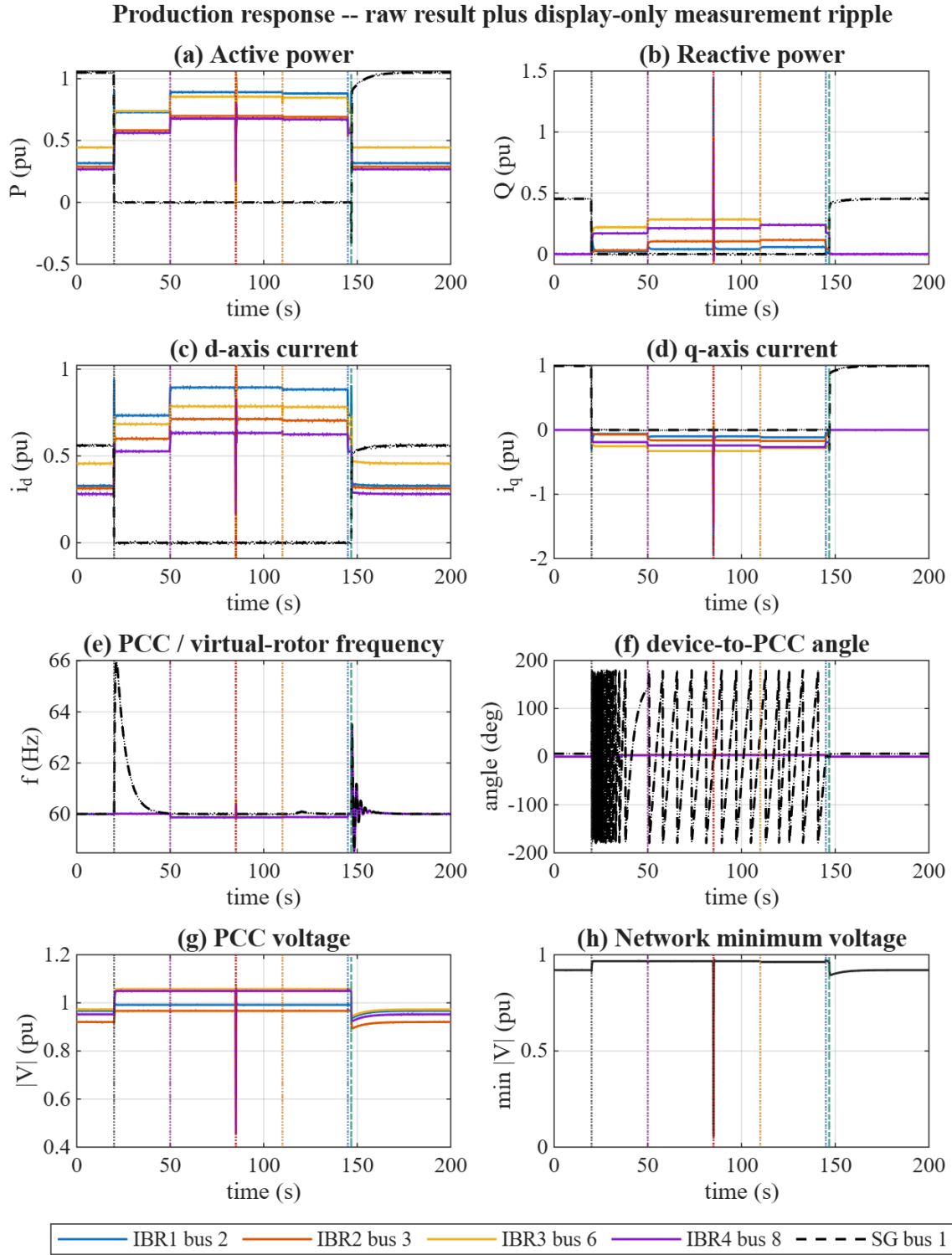


Figure 4: IBR and SG active/reactive power, frame currents, frequency, device-to-PCC angle, PCC voltage, and network minimum voltage. Solid traces are raw production results; dotted band-limited ripple is display-only and does not enter any equation or decision. Remaining vertical event excursions belong to the raw switching transient, not to the ripple.

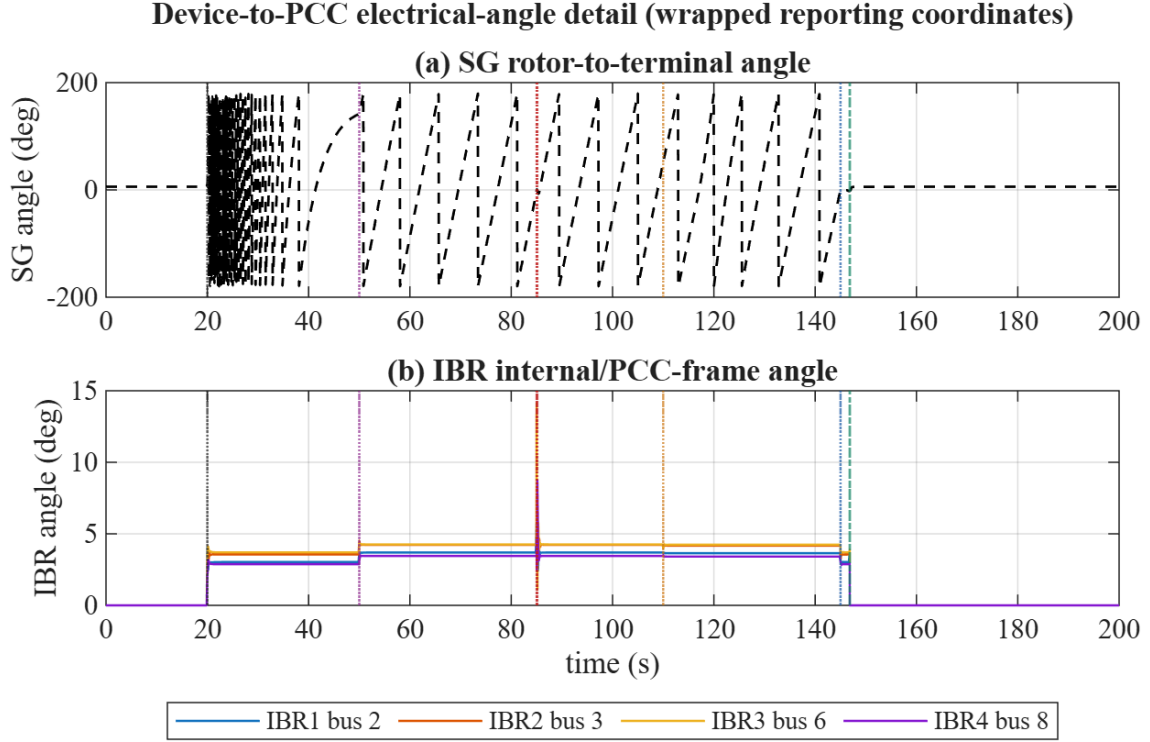


Figure 5: Angle detail. The offline SG rotor slips relative to the energized grid; the plotted coordinate is wrapped to $[-180, 180)$ degrees.

At the accepted close, the synchronism mismatch is $\Delta V = 0.0163$ pu, $\Delta f = 2.75 \times 10^{-6}$ pu, and $\Delta \theta = 3.76^\circ$; all three lie inside the guard limits (0.05 pu, 0.001 pu, 10°). At the 200 s endpoint the all-bus voltage range is 0.9203–1.0000 pu: the SG and IBR regulating buses sit inside the $\pm 5\%$ band, but five remote load buses solve below $V_{\min} = 0.95$ pu. SG electrical power is 1.0497 pu, reactive output is 0.4531 pu, and SG frequency is 60.0002 Hz, i.e. settled from the transient excursion. The trajectory therefore demonstrates a successful declared handback and frequency recovery, while the endpoint reactive profile shows the new operating point has not yet returned every bus to the $\pm 5\%$ band.

6 Evidence boundary and reproduction

The primary run uses $\Delta t = 0.0125$ s. The accepted-close time (146.825 s) is an event-landed output of the synchronism guard, not a scheduled constant, so event-time step-independence is not claimed. The run proves numerical execution of the declared chronology, atomic mode transfer, full KCL, synchronism-gated close, and return to all GFL. It does not validate the diagnostic synchronizer, governor parameters, energy reserve, or a general protection design against field measurements. The dotted display ripple is never used as model evidence; the solid raw traces remain visible.

```
pf_init_paths;
addpath(fullfile('scripts','reporting'));
generate_ieee14_switch_report_figures(reuse_cache=true);
```

All production solvers are project-owned MATLAB. The targeted numerical, producer and consumer, hybrid event, fail-closed, and external-solver-prohibition suites pass 108/108 tests. The

full repository regression was intentionally omitted under the repository risk policy because these targeted gates cover the changed producer, consumers, and failure paths.